Blood Cells, Molecules, and Diseases 42 (2009) 159–166

Contents lists available at ScienceDirect

Blood Cells, Molecules, and Diseases

journal homepage: www.elsevier.com/locate/ybcmd

Promising results of the chaperone effect caused by iminosugars and aminocyclitol derivatives on mutant glucocerebrosidases causing Gaucher disease

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ARTICLE INFO

Article history:
Submitted 8 September 2008
Revised 17 November 2008
Available online 22 January 2009

(Communicated by M. Lichtman, M.D., 18 November 2008)

Keywords:
Gaucher disease
Therapeutic strategy
Chaperones
GBA mutations
Enzyme activitiy

ABSTRACT

Gaucher disease is an autosomal recessive disorder. It is characterized by the accumulation of glucosylceramide in lysosomes of mononuclear phagocyte system, attributable to acid β-glucosidase deficiency. The main consequences of this disease are hepatosplenomegaly, skeletal lesions and, sometimes, neurological manifestations. At sub-inhibitory concentrations, several competitive inhibitors act as chemical chaperones by inducing protein stabilization and increasing enzymatic activity. Here we tested two iminosugars (NB-DNJ and NN-DNJ) and four aminocyclitols with distinct degrees of lipophilicity as pharmacological chaperones for glucocerebrosidase (GBA). We report an increase in the activity of GBA using NN-DNJ, NB-DNJ and aminocyclitol 1 in stably transfected cell lines, and an increment with NN-DNJ and aminocyclitol 4 in patient fibroblasts. These results on specific mutations validate the use of chemical chaperones as a therapeutic approach for Gaucher disease. However, the development and analysis of new compounds is required in order to find more effective therapeutic agents that are active on a broader range of mutations.

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Introduction

Lysosomal storage diseases are a group of disorders caused by the loss of function of lysosomal enzymes, which leads to the intralysosomal storage of non-degraded substrates. The disorders are classified on the basis of the substrate that is accumulated. Gaucher disease (GD, OMIM 230800) is the most prevalent sphingolipidosis caused by deficiency of glucocerebrosidase (GBA, E.C. 3.2.1.45), which produces the progressive accumulation of glucosylceramide. Clinically, GD is classified into three major types depending on the absence (Type I) or presence (Type II and III) of central nervous system involvement. The main symptoms of GD are anaemia, thrombocytopenia, hepatosplenomegaly and skeletal disease. For a review see Beutler and Grabowski [1].

Two disease-specific therapies have been approved to treat GD. Enzyme replacement therapy (ERT) has been applied for more than 15 years and has proved successful mainly for visceral and haematological symptoms of type I patients. However, since the recombinant enzyme does not cross the blood–brain barrier, its efficacy in neurological involvement is, if any, limited [2]. Moreover, it is an expensive lifetime treatment that requires frequent intravenous infusion of enzyme. The other approved treatment is substrate reduction therapy, which is based on the inhibition of glucosylceramide synthase (GCS), the rate-limiting first step in the glycosphingolipid biosynthetic pathway, by the oral administration of *N*-butyl-deoxynojirimycin (NB-DNJ) [3]. This reduction therapy is used for type I patients for whom ERT is not a therapeutic option. The small size of NB-DNJ makes it of potential use for neurological cases.

To date, other alternative strategies, such as gene therapy, have had very limited success in the treatment of GD. However, new experimental approaches in cellular and animal models have been assayed either for conventional gene therapy [4,5] or based on the partial inhibition of the GCS gene using siRNAs [6].

In the last few years, a new line of research, using small molecules that act as chemical chaperones, has emerged [7]. This approach is based on the assumption that some mutations cause the misfolding of lysosomal enzymes after their synthesis. Misfolding is responsible for enzyme degradation in the endoplasmic reticulum (ER), thereby preventing enzyme transport to the lysosome. In this scenario, the chaperone stabilizes the mutant protein, thereby allowing that, at least, some molecules reach their final destination. A number of iminosugars and aminocyclitols show a chaperone-like profile *in vitro* towards GBA [8]. However, the first molecule reported to act as a chaperone *in vivo* for GD was the iminosugar *N*-(*n*-nonyl) deoxynojirimycin (NN-DNJ) [9]. The addition of sub-inhibitory concentrations of NN-DNJ to fibroblasts of a GD patient, homozygous for the N370S mutation, resulted in a two-fold increase in GBA activity. Furthermore, several mutant GBA enzymes were shown to increase their activity following the addition of *N*-(*n*-butyl)deoxynojirimycin (NB-DNJ, miglustat, Zavesca®) to COS

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doi:10.1016/j.bcmd.2008.11.002

cells transfected with the corresponding mutant cDNAs [10]. Other iminosugars are also active on the N370S mutation [11,12] and also on several mutations that cause GD [13–15]. In addition, some non-sugar GBA inhibitors [16] and isofagomine [17] also have a chaperone effect on the N370S mutant enzyme in GD patient cells. Clinical trials using isofagomine tartrate (AT2101) are currently being conducted [18].

Here we analyzed the effect of six compounds, namely iminosugars NB-DNJ and NN-DNJ and aminocyclitols 1–4 [19] (Fig. 1), on COS cells transfected with either the wild-type or mutant *GBA* cDNAs. Ten distinct mutants were assayed. The effect of these compounds was also tested on the residual β-glucosidase activity of fibroblasts from patients with diverse genotypes.

Materials and methods

Iminosugars and aminocyclitols

NB-DNJ was purchased from Sigma. NN-DNJ was obtained from tetra-*O*-benzyldeoxynojirymycin [20] followed by reductive alkylation using described protocols [21]. Aminocycitols 1–4 were obtained as previously described [19].

Plasmid construction and mutagenesis

Wild-type and mutant GBA cDNAs (N188S, N188S;E326K, E326K, N370S, I402T, D409H and L444P) cloned in pUC18 were obtained as described in Montfort et al. [22]. Alleles bearing mutations G202R, H255Q, and G377S, and the double mutant H255Q;D409H were generated using the Quick-Change® XL Site-Directed Mutagenesis Kit (Strategene, La Jolla, California, USA) and primers containing the desired mutation. Primer information is available on request. The wild-type and all mutant cDNAs were subcloned into the expression vector pCDNA3.1 (Invitrogen, Paisley, Scotland, UK). The plasmid was purified using HiSpeed Plasmid Midi Kit (Qiagen, Hilden, Germany). All PCR products were sequenced using the BigDye Terminator v3.1 Cycle Sequencing Kit (Applied Biosystems, Foster City, California, USA).

Transient and stable GBA transfections

African green monkey kidney cells (COS-7) were grown in Dulbecco's modified Eagle's medium (DMEM) (Sigma, Deisenhofen, Germany) supplemented with 10% (v/v) bovine serum (BS) (Gibco® Invitrogen, Paisley, Scotland, UK), and 1% penicillin/streptomycin (Gibco® Invitrogen, Paisley, Scotland, UK) at 37 °C in a humidified, 5% CO_2 incubator. Cells were transfected with 6 mg of Lipofectamine® 2000 (Invitrogen, Paisley, Scotland, UK). After transfection, cells were grown for 2 days without selection.

Stable transfectants were obtained with geneticin (G418) selection (500 mg/mL). Clones were grown in DMEM, 10% BS, 1% penicillin/streptomycin, and 500 mg/mL G418/Geneticin® (Invitrogen, Paisley, Scotland, UK). The presence of the transfected gene in the selected clones was assessed by Western blot.

SDS-PAGE and Western blotting analysis

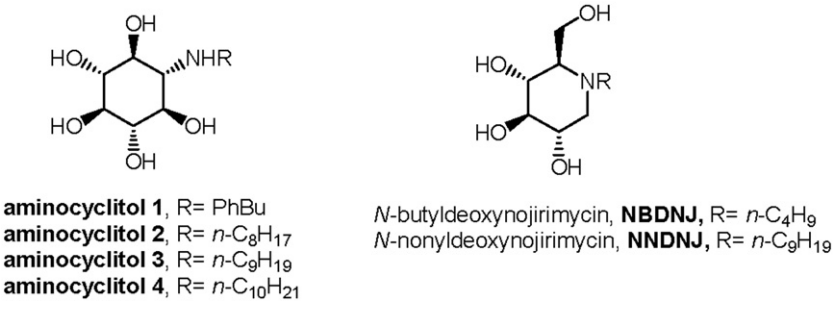
Proteins were subjected to SDS-PAGE (12% polyacrylamide) and electrophoretically transferred onto Immobilon-P Transfer nitrocellulose membranes (Millipore, Billerica, Massachusetts, USA). About 5 μg was loaded per lane and the membrane was blocked 2 h at room temperature with 5% non-fat milk in PBS, containing 0.1% Tween 20 (MTP). After incubation overnight at 4 °C with anti-GBA monoclonal antibody (Abnova® Corporation, Taipei City, Taiwan) (or anti-alpha-tubulin monoclonal antibody, Sigma, Deisenhofen, Germany) in MTP, the membrane was washed three times with PBT and three times with PBS. Membranes were then incubated with a secondary antibody sheep anti-mouse IgG Peroxidase Conjugate (Sigma, Deisenhofen, Germany) (1:10,000 in MTP) for 2 h at room temperature, followed by three washes with PBT and three washes with PBS. Detection reaction was performed by incubating the membrane in a luminol solution (100 mM Tris–HCl pH 9, coumaric acid 0.2 mM, luminol 1.25 mM and 0.3% H_2O_2) for 2 min at RT. The membrane was exposed to Amersham HyperfilmTM ECL (GE Healthcare Limited, Buckinghamshire, UK).

Inhibition studies

To establish the best range of concentrations of the compounds to be used in the chaperone assays, the inhibitory effect of the aminocyclitols and iminosugars on β-glucosidase activity was assayed in crude extract of COS-cells and fibroblasts. The analysis was performed using the 4-methylumbelliferyl β-D-glucoside substrate (4MU-Glc; 5 mM), in 0.5 M citrate buffer, pH 5.5, in the presence of Triton X-100 and taurocholate [23].

β-Glucosidase assay in lysed or intact COS-7 cells

Transient transfected cells were washed once in PBS, trypsinized, harvested by centrifugation and stored at −80 °C. β-glucosidase activity was measured with 4MU-Glc as described [23]. Stable COS-7 transfectants were cultured in 24-well assay plates during 6 days in DMEM, 10% BS at 37 °C under 5% CO_2 either with or without NB-DNJ, NN-DNJ, or aminocyclitols 1–4 (see Fig. 1). Culture media was replaced every 3 days with fresh media supplemented with the corresponding compound, and cells were washed, harvested and stored as indicated above. Cell pellets were resuspended in 100 ml of PBS. Protein levels were determined by the Bradford method, using bovine serum albumin standards (BSA, Sigma, Deisenhofen, Germany) and Bio-Rad Protein Assay (Bio-Rad Laboratories, Hercules, California, USA). Cells were sonicated and the levels of acid β-glucosidase activity were monitored by enzyme assay using 4MU-Glc, following the protocol by Sawkar et al. [9]. The amount of 4-methylumbelliferone was analyzed with a fluorometer Spectramax Gemini XPS (Molecular Devices Corporation; excitation wavelength 355 nm, emission wavelength 460 nm). The ratio of enzyme activity with/without aminocyclitol/iminosugar was calculated. The values shown are the results of at least two experiments with 3 or 4 replicates. For intact cells, the protocol



aminocyclitol 1, R= PhBu
aminocyclitol 2, R= *n*-C_8H_{17}
aminocyclitol 3, R= *n*-C_9H_{19}
aminocyclitol 4, R= *n*-$C_{10}H_{21}$

N-butyldeoxynojirimycin, **NBDNJ,** R= *n*-C_4H_9
N-nonyldeoxynojirimycin, **NNDNJ,** R= *n*-C_9H_{19}

Fig. 1. Iminosugars (NN-DNJ and NB-DNJ) and aminocyclitols 1–4 used in this study.

was similar to the one described above except that the reaction was performed directly in the plate well, without trypsinizing and harvesting the cells.

Fibroblast culture assays

Skin fibroblasts were obtained from non-neurological (type 1) and neurological (types 2 and 3) GD patients with distinct genotypes, which were previously diagnosed at the *Institut de Bioquímica Clínica*. Fibroblasts from healthy individuals were used as controls. Fibroblast cultures were established following routine procedures in Eagle's minimal essential medium (MEM) with 15% inactivated fetal calf serum. The cells used were between the 3rd and the 10th passage.

For the β-glucosidase assay in lysed cells, 200,000 cells were placed in flat tubes (10 cm2) with MEM containing the aminocyclitol or the iminosugar dissolved in DMSO. For each experiment, four to five flat tubes with the compound at concentrations ranging between 2.5 and 60 μM were set up. Untreated and treated cells were incubated for 4 days at 37 °C. When incubations were longer, media were replaced after 3 days. Prior to the β-glucosidase activity assay, media supplemented with aminocyclitols 1–4, NN-DNJ or NB-DNJ were removed, and cells were rinsed with PBS and incubated in routine culture medium for 15 min at 37 °C in order to assure complete removal of the compound. Cells were rinsed again twice with PBS and harvested after trypsin treatment. After centrifugation, the cell pellets were resuspended in water, lysed by sonication and used for assaying β-glucosidase activity, β-hexosaminidase activity (as control lysosomal enzyme), and total protein. The β-glucosidase activity assay was performed with substrate (4MU-Glc; 20 mM). Cells were incubated for 1 h at 37 °C, the enzyme reaction was stopped with 3 ml of sodium carbonate–bicarbonate buffer (pH 10.7), and fluorescence was measured. β-hexosaminidase activity was measured with the fluorogenic 4-methylumbelliferyl derivative and protein content estimated following Lowry [24]. The effect of iminosugars or aminocyclitols as potential chaperones was evaluated in triplicate for each experiment.

For the assay of acid β-glucosidase in intact cells, these were plated into 24-well assay plates with MEM containing several concentrations of the aminocyclitol product or NN-DNJ or NB-DNJ and incubated as described above. Cells were washed and the enzyme assay performed as follows: substrate (100 μl, 5 mM 4MU-Glc) in 0.1 M acetate buffer (pH 4.0) was added to each well up to a total volume of 260 μl; incubation was for 1 h at 37 °C. Enzyme reaction was stopped with 2 ml of carbonate–bicarbonate buffer (pH 10.7) and the fluorescence released was measured. For each experiment untreated (no compound added) and treated cells were plated in quadruplicate.

The non-specific β-glucosidase activity was evaluated by addition of conduritol B-epoxide to control wells and was shown to account for about 1% of the activity in control fibroblasts.

Statistical analysis

For all measures, the hypothesis of normality was rejected and non-parametric Mann–Whitney *U* test was used. Normal distribution was assessed by Kolmogorov–Smirnov test. SPSS statistical programme was used for the statistical analysis. *P* values of ≤0.05 were considered statistically significant.

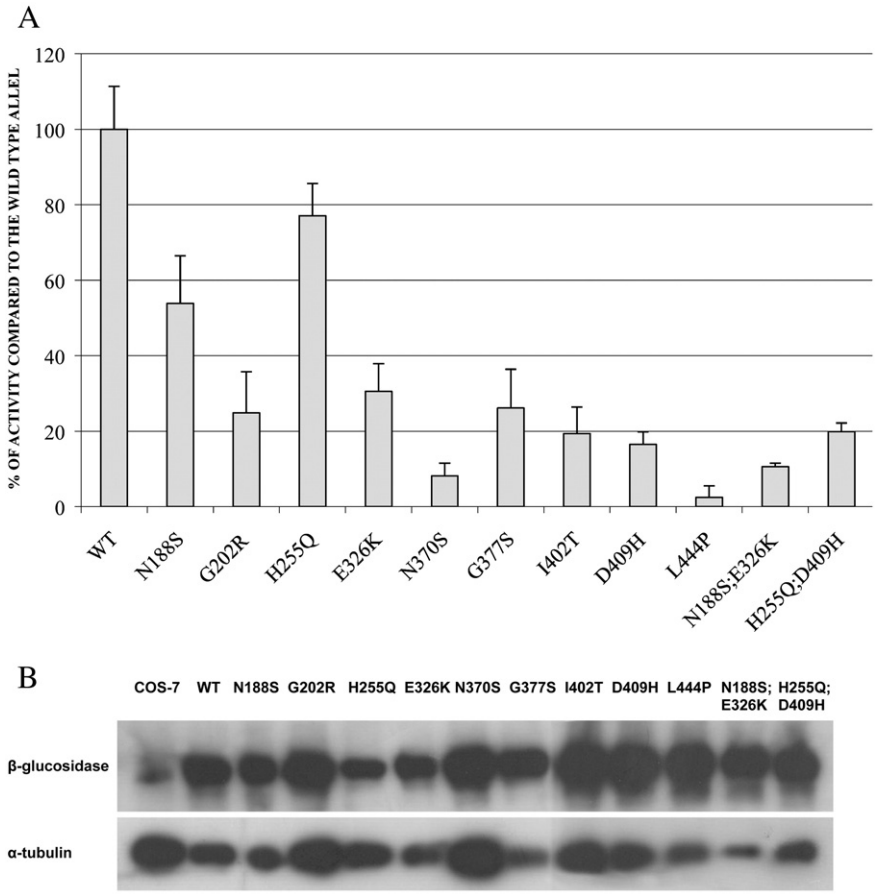


Fig. 2. Characterization of mutant glucocerebrosidases expressed in COS cells. (A) Mean activity, normalized by mg of protein, and SE for the mutant proteins, expressed as a percentage of that of wild-type enzyme. (B) Western blot analysis of the mutant proteins. α-tubulin was used as a reference loading control.

Results

Enzyme activity of mutant GBAs

COS-7 cells were transiently transfected with wild-type *GBA* cDNA and cDNAs bearing the following mutations: N188S, G202R, H255Q, E326K, N370S, G377S, I402T, D409H and L444P, and the double mutants N188S;E326K and H255Q;D409H. The activities of the enzymes produced by these constructs are shown in Fig. 2A. The N188S, H255Q, and E326K changes showed the highest residual activity: 53.84% (SD: 12.59), 77.09% (SD: 8.58), and 30.51% (SD: 7.34) of wild-type, respectively. To further characterize the mutant products, immunoblots of the normal and mutant proteins were performed. A unique band pattern, similar to that of the wild-type, was detected using monoclonal anti-human acid β-glucosidase for all the mutant enzymes, albeit with some differences in intensity (Fig. 2B).

Effect of treatment with NN-DNJ or NB-DNJ on enzyme activity in transfected cells

Iminosugars NN-DNJ and NB-DNJ (Fig. 1), previously described as having a chaperone effect on specific *GBA* mutations, were tested on mutations described in our series of patients. COS-7 cells stably transfected with constructs bearing the distinct mutations were treated with 2.5 and 5 μM of NN-DNJ for six days and the GBA activity was determined and compared with GBA activity in cells treated only with vehicle. Since we were not able to obtain stably transfected cells bearing the H255Q mutation alone, the effects of compounds on this mutation could not be analysed. Results using lysed cells for the enzyme activity assay are shown (Fig. 3A). Treatment with 2.5 μM NN-DNJ led to a 1.4-fold increase in the activity of the N188S mutant enzyme. For G377S, the double mutant N188S;E326K and the wild-type enzymes, an increase in activity

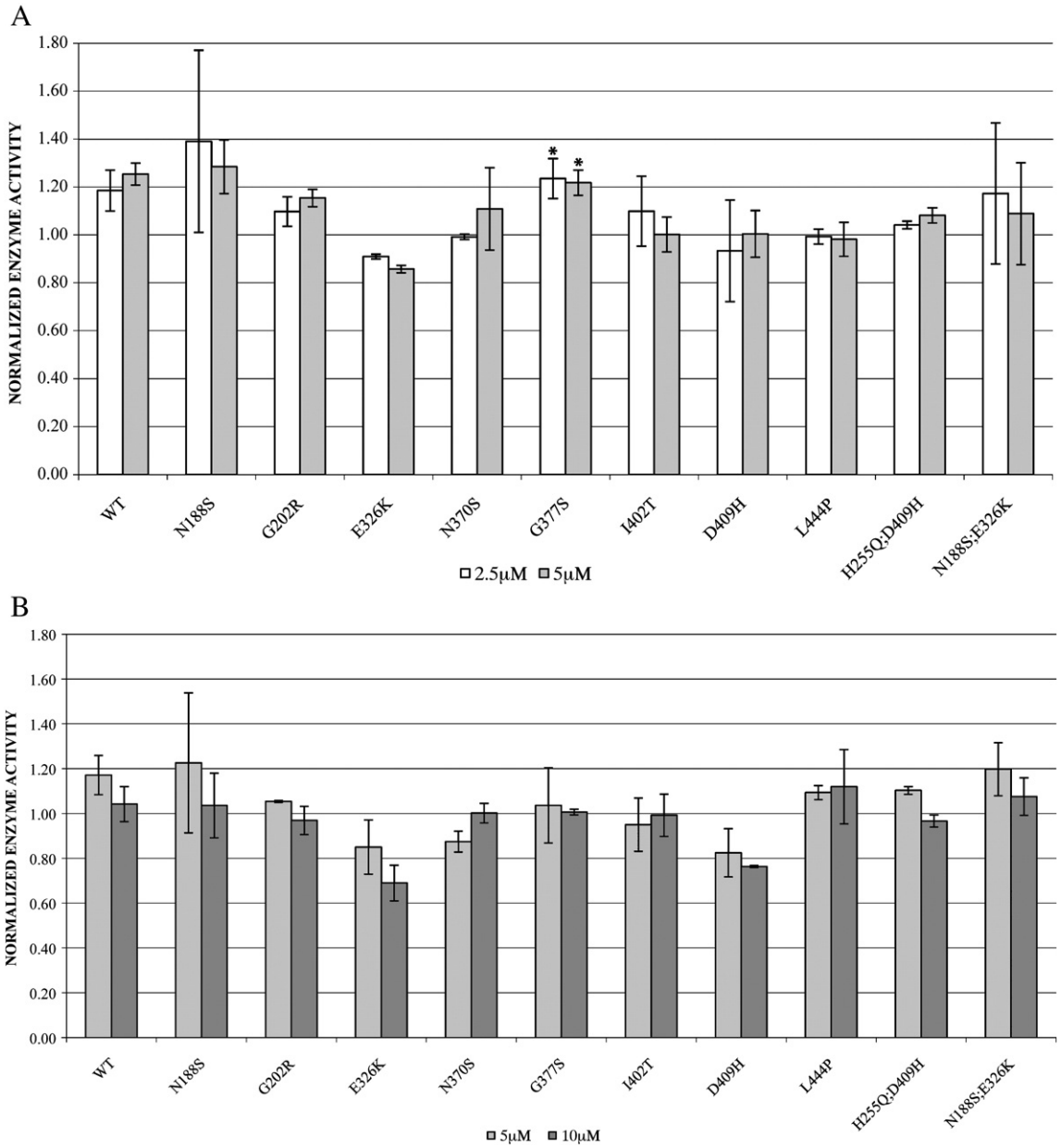


Fig. 3. Normalized β-glucosidase activity in COS-7 cells after the treatment with iminosugars for 6 days. (A) Mean and SE values after treatment with NN-DNJ. (B) Mean and SE values after treatment with NB-DNJ. The normalized enzyme activity is calculated by the ratio between treated cells and untreated cells. * $p \leq 0.05$.

(about 1.2-fold) was also observed. This enhancement in activity was also detected for these enzymes when treated with 5 μM NN-DNJ, although to a lesser extent in the cases of N188S and the double mutant N188S;E326K. Smaller increases in enzyme activity occurred for other mutant alleles such as G202R, N370S and I402T, either at one or both concentrations. For E326K a small decrease in activity was observed. No change in activity was found for the rest of the mutant enzymes.

Iminosugar NB-DNJ was tested at 5 and 10 μM. A 20% increase in activity was observed for N188S and the double mutant N188S;E326K when treated with 5 μM (but not 10 μM) NB-DNJ (Fig. 3B). The treated wild-type enzyme showed a similar increase. It should be noted that these enzymes were also activated by NN-DNJ. However, G377S activity, which was positively affected by NN-DNJ treatment, showed no increase when treated with NB-DNJ. As with NN-DNJ, a decrease in E326K β-glucosidase activity was also observed with NB-DNJ. A similar negative effect was detected for D409H. No other significant changes in activity were detected for the rest of the mutant proteins, except for a slight increase for L444P and the double mutant H255Q; D409H. For most mutant enzymes, assays with intact cells were also performed and similar results were obtained (not shown).

Effect of treatment with aminocyclitols on enzyme activity in transfected COS cells

Aminocyclitols 1–4 (Fig. 1) were also tested at a range of concentrations in COS-7 cells stably transfected for 6 days with constructs bearing the distinct mutations. While for linear *N*-alkyl aminocyclitol derivatives 2–4 no effect was detected on any of the enzymes studied, a positive effect was observed for aminocyclitol 1 on some of the mutant enzymes (Fig. 4). A 1.3- to 1.4-fold increase in activity was observed for the wild-type and N188S enzymes, at 15 and 20 μM concentrations. The double mutant N188S;E326K GBA displayed a 1.2-fold increase in activity at 15 μM. At a concentration of 20 μM, N370S showed a 15% increase in activation. The effect of the compound on the activities of the other GBA mutations was either null or negative. As for the above iminosugars, intact cell assays were also performed and comparable results were obtained (not shown).

Effect of treatment with iminosugars and aminocyclitols on β-glucosidase activity in fibroblasts from GD patients

The compounds were also tested on the residual enzyme activity of fibroblasts from GD patients with distinct genotypes, which have been previously characterized [25]. The genotypes were: N370S/N370S, N370S/L444P, L444P/L444P, D409H/N188S;E326K, D409H/D409H, I402T/I402T, L444P/G202R, D409H/L444P;E326K, and L444P;E326K/G202R. Results using lysed cells for enzyme activity assay are shown (Figs. 5 and 6). Treatment with NN-DNJ caused a remarkable increase in enzyme activity of fibroblasts with genotypes N370S/N370S, D409H/N188S;E326K, and N370S/L444P (Fig. 5). For the N370S/N370S enzyme, a 1.7-fold increase was observed at low concentrations. Using intact cells, a 1.9-fold increase was observed upon addition of 5 and 10 μM NN-DNJ to the culture media. At higher concentrations (20–60 μM NN-DNJ), a sustained increase in enzyme activity (1.8-fold) was still observed (results not shown). Cells from patients with genotype N370S/L444P showed a moderate increase in activity (1.2 to 1.4-fold) at low NN-DNJ concentrations (2.5–10μM). For D409H/N188S;E326K fibroblasts, obtained from a patient with GD type I, the increase was 1.2- to 1.4-fold, at concentrations of 5 and 10 μM. No effect was observed on the residual enzyme activity of the related genotype D409H/L444P;E326K, identified in a patient with GD type III [26]. Smaller effects were observed for fibroblasts bearing the L444P;E326K/G202R and L444P/G202R genotypes (not shown). NN-DNJ did not affect β-glucosidase activity in lysed cells bearing genotypes homozygous for mutation D409H (Fig. 5), L444P or I402T (not shown).

Treatment with NB-DNJ produced minor increases in enzyme activity (about 1.2-fold) in some cases, similarly to that observed in lysed wild-type cells. In contrast, a clear inhibition of enzyme activity was observed in lysed cells with genotypes D409H/D409H, D409H/L444P;E326K (but no effect on D409H/N188S;E326K) and I402T/I402T in the 5 to 60 μM NB-DNJ concentration range (results not shown).

Fibroblasts from GD patients with distinct genotypes were treated with aminocyclitols 1–4. A positive response was obtained only for those bearing the L444P/G202R and L444P;E326K/G202R genotypes after treatment with compound 4 (Fig. 6). A 1.5- to 1.7-fold increase in β-glucosidase activity was reached in the L444P/G202R and L444P;

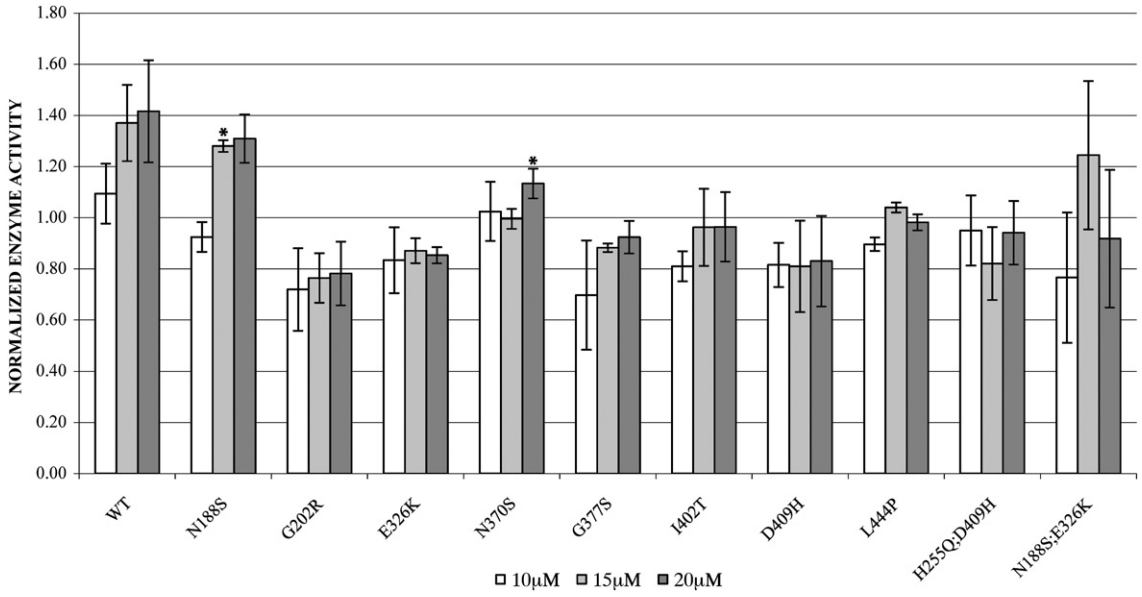


Fig. 4. Normalized β-glucosidase activity in COS-7 cells after the treatment with aminocyclitol 1 for 6 days. Mean values and SE are indicated. The normalized enzyme activity is calculated by the ratio between treated cells and untreated cells. * $p \leq 0.05$.

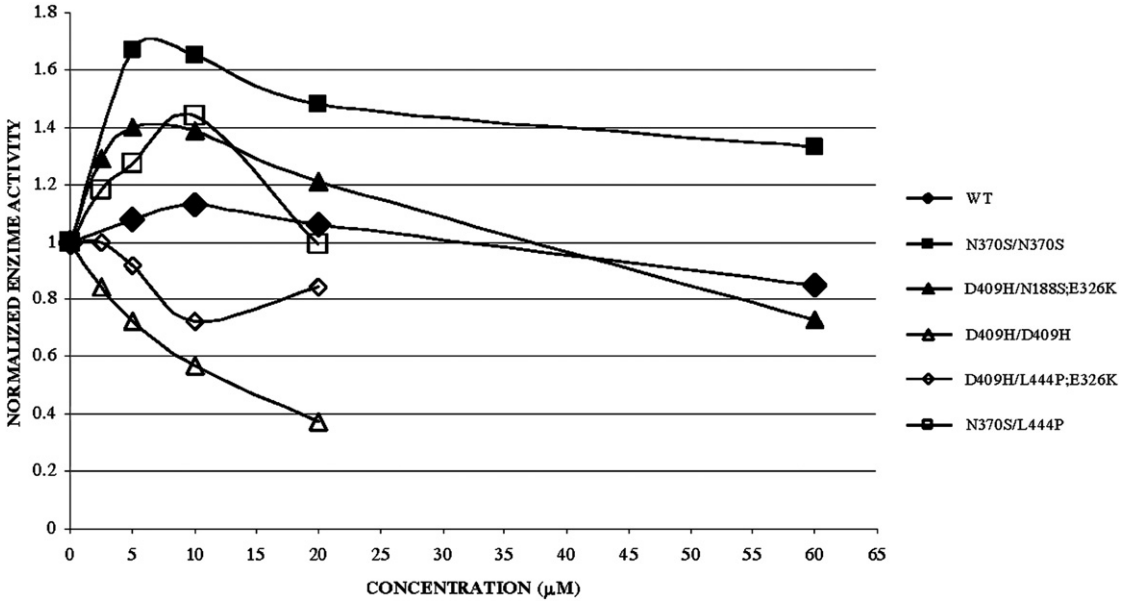


Fig. 5. Effect of NN-DNJ on β-glucosidase activity of fibroblast lines. Fibroblasts (from healthy individuals and from GD patients) were cultured for 4 days with NN-DNJ and lysed cells were assayed for β-glucosidase activity. Enzyme activity was normalized to untreated cells, which were assigned a value of 1. Means of three separate experiments are presented (SE <20%).

E326K/G202R fibroblast-lysates at 20–30 μM concentrations. In fact, a moderate enhancement of enzyme activity was observed for compound 4 at all the concentrations tested. In contrast, activity inhibition up to 60% was detected in the N370S/N370S lysed cells. Assays with intact fibroblasts grown in the presence of compound 4 were also performed and similar results were obtained. The other aminocyclitols assayed did not affect or, even, inhibited β-glucosidase (results not shown).

Discussion

Iminosugar inhibitors have been used to treat glycosphingolipidoses, such as Fabry disease [27,28], gangliosidosis GM1 [29], and gangliosidosis GM2 [30], in cellular and mouse models. For GD, iminosugars NN-DNJ [9] and NB-DNJ [10] have been tested in fibroblasts from GD patients and in stably transfected COS-7 cell lines, respectively.

Here we tested NB-DNJ and NN-DNJ and aminocyclitols 1–4, which show distinct degrees of lipophilicity, on COS-7 cells stably transfected either with the wild-type GBA cDNA or with cDNAs bearing the mutations N188S, G202R, E326K, N370S, G377S, I402T, D409H, L444P, N188S;E326K, and H255Q;D409H. Moreover, we also analysed fibroblasts from GD patients with distinct genotypes, including the above-mentioned mutations. We found an increase in β-glucosidase activity induced by some of the compounds tested on specific mutations.

NN-DNJ was the first to be reported to have chaperone effect on wild-type and the N370S GBA enzymes in fibroblasts from GD patients [9]. In agreement with these findings, we reproduced these results in fibroblasts with the N370S/N370S genotype, and we also found a positive effect on N370S/L444P and D409H/N188S;E326K fibroblasts, in the latter case probably increasing the residual enzymatic activity derived from the N188S allele. This finding was consistent with results in COS cells transfected with the N188S mutation. In these cells, the effect on the N370S enzyme was lower, while the wild-type enzyme and the mutant G377S exhibited a significant increment of activity over untreated cells.

Iminosugar NB-DNJ had not previously been assayed in fibroblasts. Our results showed no significant enhancement of enzyme activity in any patient fibroblasts analyzed, with a clear inhibition of the D409H/D409H and I402T/I402T mutant enzyme activities. In contrast, a clear stimulation was observed using COS cells transfected with the N188S and N188S;E326K cDNAs. A moderate increase in enzymatic activity was detected for the wild-type allele, in fibroblasts and COS cells. Alfonso et al. [10], working on COS cells, reported stimulation of the wild-type and some mutant enzymes, most of them differing from those studied here. Mutation N370S was common to both studies, but while Alfonso et al. found a clear stimulation (although with a very large SD), we did not observe any clear effect.

Aminocyclitols 2 and 3 did not show any chaperone effect either on fibroblasts from GD patients or in transfected COS cells. However, a positive effect was observed for aminocyclitol 4 in fibroblasts bearing the L444P/G202R and L444P;E326K/G202R genotypes. This increase

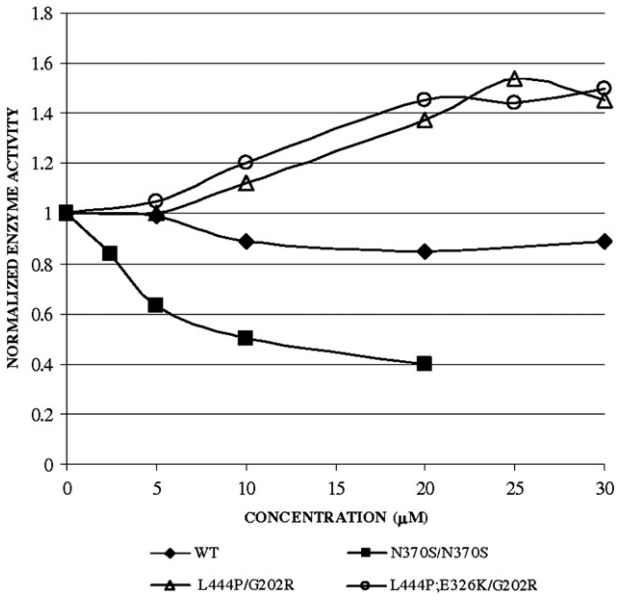


Fig. 6. Effect of the aminocyclitol 4 on β-glucosidase activity of fibroblast lines. The compound was added to the culture media for 4 days and β-glucosidase activity was measured in lysed cells. Enzyme activity was normalized to untreated cells, which were assigned a value of 1. Means of three separate experiments are presented (SE <20%).

might derive from the G202R mutant enzyme since no effect was observed for other genotypes with the L444P mutation. In contrast, the effect of this compound on COS cells was mainly inhibitory at all the concentrations assayed, including lower concentrations such as 2.5 and 5 μM. In contrast, aminocyclitol 1 stimulated the wild-type enzyme and those bearing the N188S, and N188S;E326K mutations in COS cells, while it consistently acted as an inhibitor when assayed on patient fibroblasts.

Most of the previous studies on the chaperone effect of different compounds were performed either in patients' fibroblasts [9, 11–17] or in transfected cells [10], but not in both. Only Lei et al. [31] performed the assays in both systems. The different effect for some of the compounds on transfected COS cells and patients' fibroblasts found in the present study makes it necessary to further investigate this issue and to take the results of this and other studies with caution, until the different responses of different cells lines be understood. It is difficult to speculate on the causes of these differences, but the cell type, the genetic background, the gene copy number or the effect of the endogenous enzyme of the COS cells could be partially responsible for the discrepancies. The fact that many of the fibroblasts are compound heterozygotes, adds the effect of the other allele as another variable to be considered. This is of particular interest due to the differential responses of the different mutations. In this regard, the homogeneous background of COS cells makes this heterologous expression system a good one, in spite of the limitations mentioned above.

A clear positive effect was observed for mutations that conserved a relatively high residual activity, such as N188S and E326K, and the double mutant containing them, in COS cells and fibroblasts. This observation is consistent with the expected chaperone effect, since chemical chaperones act on enzymes that retain some degree of activity. These results could be considered proof of concept for this therapeutic approach. However, for the most severely affected enzymes, the synthesis and assay of more effective compounds is required. In this regard, the long-term results of the clinical trials with isofagomine (IFG), and the pre-clinical studies that are currently being performed for several IFG-derived compounds [32] will be of interest.

The strategy followed to select the compounds tested in our study was based on the general idea that effective competitive inhibitors are more likely to show a potent chaperone effect. The compounds developed for the present study showed good inhibitory properties (IC_{50} between 1.8 and 27.6 μM) and the most active inhibitors showed a correlation with lipophilicity and, thus, with the length of the alkyl chain [33]. Moreover, they showed no toxicity either in fibroblasts or in COS cells at the range of concentrations tested. However, we achieved only partial and suboptimal results. These findings are consistent with the assumption that the most effective chaperones are not necessarily the best inhibitors, as shown also by Lei et al. [31], since the enhancement of activity could arise from more than one mechanism of action [32] or be influenced by other compound properties (membrane permeability, metabolism, enzyme selectivity or intracellular localization, among others), which may be relevant for the desired activity enhancement effects on cellular GBA enzymes.

Crystal structure analyses show that NB-DNJ and NN-DNJ bind at the active site of the GBA enzyme [34]. To date, no direct structural information on the binding of the aminocyclitols 1–4 to GBA is available, although they display substrate competitive kinetics *in vitro* [33] and, therefore, it is assumed that these compounds bind to the GBA catalytic site. Interestingly, the aminocyclitol 4 is active on fibroblasts bearing mutations L444P and/or E326K, which have been reported to be located in the GBA–SAP interaction region [35]. This observation points to a second site of action for this compound, apart from the catalytic site. This information might be useful to study and predict the effects of the different compounds. Current studies addressed at the synthesis of new aminocyclitols with improved physico-chemical properties are currently underway.

Acknowledgments

The authors thank Dr. Amadeu Llebaria and Dr. Gemma Fabriás (from RUBAM, CSIC), and Dr. Antonio Delgado (from Universitat de Barcelona and RUBAM-CSIC) for helpful discussion of the MS and the project, Anna Yudego for technical help, and Robin Rycroft for revising the English. Partial financial support from the Spanish Ministerio de Educación y Ciencia (SAF2006-12276), Ministerio de Sanidad y Consumo (FIS PI042350, PI040767 and PI040867), Ministerio de Ciencia y Tecnologia (Project MCYT CTQ2005-00175/BQU), Fondos Feder (EU), and from the Generalitat de Catalunya (SGR2005-00848) are acknowledged. The CIBER de Enfermedades Raras is an initiative of the ISCIII.

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